

Sequential anytime-valid inference (SAVI) using e-processes

Part I

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Overview

1. Motivation: peeking at p-values (Section 7.1)
2. Warm-up: Wald's SPRT (Section 7.2)
3. Core definitions for validity (Section 7.3)
4. Ville & optional stopping (Section 7.4)
5. Core definitions for asymptotic growth (Section 7.3)
6. Building e-processes (Section 7.5)
7. Testing as betting (Section 7.6)

Outline

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The problem with peeking at p-values

- A scientist computes a running p-value P_n after each new observation and stops at first time $\tau = \inf\{n \geq 1 : P_n \leq \alpha\}$.
- The final sample size (the stopping time) τ is random.
- For each fixed n , $\mathbb{P}(P_n \leq \alpha) \leq \alpha$ under H_0 , but typically $\mathbb{P}(P_\tau \leq \alpha) > \alpha$.
- Root cause: P_n is valid for a **fixed** horizon, not all random stopping times.
- Even under $\mathcal{H}_0$, under most typical cases, τ is almost surely finite.
- $\implies$ One will eventually stop and reject the null!

Anytime-valid goal

Construct evidence processes that remain valid at **all** times and for **any** stopping rule.

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Wald's Sequential Probability Ratio Test (SPRT)

- Testing simple $H_0 : \mathbb{P}$ vs $H_1 : \mathbb{Q}$.
- $L_t = \prod_{i=1}^t \frac{d\mathbb{Q}}{d\mathbb{P}}(X_i)$.
- LRT: **fixed** sample size test.
 - Optimal by Neyman-Pearson lemma.
- SPRT: sampling the data one by one, **continually updating** L_t .
 - Stop when $L_t \notin (\gamma_0, \gamma_1)$ and accept H_0 if $L_t \leq \gamma_0$, reject if $L_t \geq \gamma_1$.
 - Thresholds chosen by α, β .
 - e.g., $\gamma_0 = \beta$, $\gamma_1 = 1/\alpha$ suffices to keep the type-I and type-II errors below α and β respectively.

Remark

L_t is an e-process & test martingale under $\mathbb{P}$.

SPRT versus LRT

Data-generating distribution Setting	$p = 0.5$ null	$p = 0.55$ misspecified	$p = 0.6$ alternative	$p = 0.65$ misspecified
SPRT expected sample size	137.7	234.5	139.3	76.1
LRT pre-specified sample size	280	280	280	280
Power of SPRT	0.04	0.52	0.96	1.00
Power of LRT	0.05	0.48	0.95	1.00

Table 7.1: A comparison of SPRT and LRT for testing $p = 0.5$ against $p = 0.6$, with target type-I error rate at 0.05 and target type-II error rate at 0.05. All numbers are produced with Monte Carlo simulations using 10,000 repetitions.

Key takeaway

SPRT achieves optimal sample size and error tradeoffs.

Wald vs. SAVI: Key Differences

- **Decisions vs. Evidence:**

- Wald focused on quick binary decisions with fixed error rates;
- SAVI emphasizes sequentially updated *evidence measures*. Decision-making is secondary, though SAVI rules can resemble Wald's.

- **Single vs. All Stopping Times:**

- Wald used one stopping rule;
- SAVI defines an *anytime-valid e-process* with guarantees at every stopping time, allowing continuous evidence monitoring.

- **Optimality:**

- Wald optimized Type I/II error trade-offs;
- SAVI uses *log-optimality* (max expected log-evidence).
- Both coincide for simple cases via the likelihood ratio martingale.

- **Beyond Likelihood Ratios:**

- SPRT relies on likelihood ratios;
- SAVI extends to broader settings, including composite hypotheses.

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Setup: filtrations, processes and stopping times

Filtered probability space

- Filtration $\mathbb{F} = (\mathcal{F}_t)_{t \in \mathcal{T}}$, with $\mathcal{T} = \mathbb{N}_0$ or $\{0, \dots, n\}$.
- $\mathcal{F}_\infty = \sigma(\cup_{t \in \mathcal{T}} \mathcal{F}_t)$, $\mathcal{T}_+ = \mathcal{T} \setminus \{0\}$.
- Usually $\mathcal{F}_t = \sigma(X_1, \dots, X_t)$, $\mathcal{F}_0$ trivial or independent.

Processes

- $Y = (Y_t)$ is adapted if Y_t is $\mathcal{F}_t$ -measurable.
- Predictable if Y_t is $\mathcal{F}_{t-1}$ -measurable ($t \geq 1$).

Stopping times

- τ : integer-valued r.v. with $\{\tau \leq t\} \in \mathcal{F}_t$.
- We can't peak into the future.
- Set of all stopping times: $\mathbb{T}$.

Setup: sequential tests

Sequential tests for $\mathcal{P}$

- Adapted binary process $\varphi = (\varphi_t)$; reject when $\varphi_t = 1$.
- Level- α : $\sup_{\mathbb{P} \in \mathcal{P}} \mathbb{P}(\exists t \geq 1 : \varphi_t = 1) \leq \alpha$. (7.1)

- WLOG φ_t nondecreasing; let $\tau = \inf\{t : \varphi_t = 1\}$.
- Equivalent: $\mathbb{E}^{\mathbb{P}}[\varphi_\tau] \leq \alpha, \forall \mathbb{P} \in \mathcal{P}, \tau \in \mathbb{T}$. (7.2)

Martingales

- An integrable process (M_t) is a (super)martingale under $\mathbb{P}$ if

$$\mathbb{E}^{\mathbb{P}}[M_t \mid \mathcal{F}_{t-1}] \leq M_{t-1}. \quad (7.3)$$

Test (super)martingales and e-processes

Definition (Test (super)martingale; e-process)

(i) $M = (M_t)_{t \geq 0}$ is a *test (super)martingale* for $\mathcal{P}$ if for every $\mathbb{P} \in \mathcal{P}$:

1. $M_t \geq 0$ $\mathbb{P}$ -a.s.,
2. M is a (super)martingale under $\mathbb{P}$,
3. $\mathbb{E}^{\mathbb{P}}[M_0] \leq 1$.

(ii) $E = (E_t)_{t \geq 0}$ is an *e-process* for $\mathcal{P}$ if it is adapted and either (equivalently):

1. $\mathbb{E}^{\mathbb{P}}[E_\tau] \leq 1$ for all stopping times τ and $\mathbb{P} \in \mathcal{P}$; or
2. $\exists$ test martingale family $(M^{\mathbb{P}})_{\mathbb{P} \in \mathcal{P}}$ with $E_t \leq M_t^{\mathbb{P}}$ $\mathbb{P}$ -a.s. for all t and all $\mathbb{P} \in \mathcal{P}$.

Optional stopping theorem

Fact (Optional stopping for nonnegative supermartingales)

If M is a test supermartingale for $\mathbb{P}$, then for any stopping time τ and $\sigma \leq \tau$,

$$\mathbb{E}^{\mathbb{P}}[M_{\tau} \mid \mathcal{F}_{\sigma}] \leq M_{\sigma}.$$

In particular, by taking $\sigma = 0$, we have

$$\mathbb{E}^{\mathbb{P}}[M_{\tau}] \leq 1. \quad (\text{tower property} + \text{initializing at } 1)$$

If M is an e-process for $\mathcal{P}$, then $\mathbb{E}^{\mathbb{P}}[M_{\tau}] \leq 1$ for all $\mathbb{P} \in \mathcal{P}$.

Remark

Every e-process remains an e-variable at *any* stopping time.

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Ville's inequality $\Rightarrow$ sequential tests by thresholding

E-processes can be converted to sequential tests using Ville's inequality.

Fact (Ville's inequality)

If M is an e-process for $\mathcal{P}$, then for all $\alpha \in [0, 1]$ (should be $(0, 1]$),

$$\mathbb{P}(\exists t \in \mathbb{N} : M_t \geq \frac{1}{\alpha}) \leq \alpha, \quad \forall \mathbb{P} \in \mathcal{P}$$

$$\iff \sup_{\mathbb{P} \in \mathcal{P}} \mathbb{P}\left(\sup_{t \geq 0} M_t \geq 1/\alpha\right) \leq \alpha,$$

$$\iff \sup_{\mathbb{P} \in \mathcal{P}, \tau \geq 0} \mathbb{P}(M_\tau \geq 1/\alpha) \leq \alpha,$$

Remark

For any fixed t , M_t is an e-variable. Ville's inequality is best seen as a time-uniform generalization of Markov's inequality that applies to e-processes rather than e-variables.

Ville's inequality $\Rightarrow$ sequential tests by thresholding

- $\phi_t = \mathbf{1}\{M_t \geq 1/\alpha\}$ defines a **level- α sequential test**.

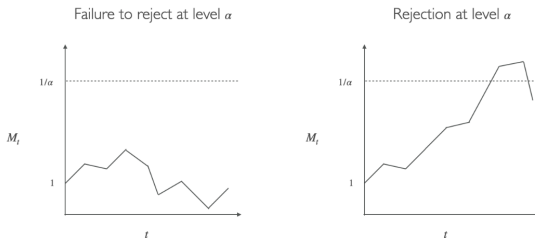


Figure 7.7: A visualization of sequential testing with Ville's inequality applied to an e-process M .

Proposition (Every sequential test comes from an e-process)

Every level- α sequential test φ for $\mathcal{P}$ can be obtained by thresholding some e-process M for $\mathcal{P}$ at $1/\alpha$; i.e., for any level- α sequential test ϕ , there exists an e-process M such that

$$\phi_t = \mathbb{1}\{M_t \geq 1/\alpha\} \quad \text{is level-}\alpha.$$

Ville's inequality $\Rightarrow$ sequential tests by thresholding

Proof.

- Define $M_t = \phi_t/\alpha$. Then $M_t \in \{0, 1/\alpha\}$, and $M_t \geq 1/\alpha$ iff $\phi_t = 1$.
- M is an e-process for $\mathcal{P}$:

$$\mathbb{E}^{\mathbb{P}}[M_\tau] = \mathbb{E}^{\mathbb{P}}[\phi_\tau/\alpha] \leq 1, \quad \forall \mathbb{P} \in \mathcal{P}, \quad \forall \tau \in \mathbb{T}.$$

by the definition of a sequential test ϕ at level α .

- Thus M satisfies the e-process property, completing the proof.

Optional Continuation (Section 4.3)

If M^A and M^B are e-processes for $\mathcal{P}$ and τ being a stopping time, such that $M_t^B = 1$ on $\{t \leq \tau\}$. Define the process $\tilde{M}$ by

$$\tilde{M}_t := \begin{cases} M_t^A, & t \leq \tau, \\ M_\tau^A M_{t-\tau}^B, & t > \tau, \end{cases}$$

for any $\mathbb{F}$ -stopping time τ .

- **Interpretation:** You can stop an e-process at any data-dependent time, then restart a new valid one by multiplying the evidence so far with new evidence. The combined process $\tilde{M}$ remains an e-process for $\mathcal{P}$.
- **Proof:** Splitting the expectation at τ and using the tower property together with the e-process property of M^A and M^B to show that $\tilde{M}$ still satisfies the test supermartingale property.

From e-process to p-process

Definition (p-process)

A nonnegative process $P = (P_t)_{t \in \mathbb{T}}$ is called a **p-process** if P_τ is a p-variable for every stopping time $\tau \in \mathcal{T}$ under every $\mathbb{P} \in \mathcal{P}$.

- If M is an **e-process**, then $P_t := \inf_{1 \leq s \leq t} \frac{1}{M_s}$ is a **p-process**.
- M_t (the e-process) reflects the **current level of evidence** — how strongly the data so far support rejecting the null at time t .
- The corresponding P_t (constructed as $P_t = \inf_{s \leq t} \frac{1}{M_s}$) instead reflects the **strongest evidence ever observed up to now** — the best historical point in time where the p-value was smallest (i.e., M_s was largest).

Takeaway

An **e-process** measures evidence based on what we currently have, while a **p-process** measures evidence based on the **best** evidence we ever accumulated in the past.

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Asymptotic growth

Definition (Asymptotic growth rate and log-optimality)

For alternative $\mathbb{Q}$, an e-process E is **consistent** against $\mathbb{Q}$ if $E_t \rightarrow \infty$, $\mathbb{Q}$ almost surely as $t \rightarrow \infty$.

The lower limit

$$\liminf_{t \rightarrow \infty} \frac{1}{t} \log E_t$$

is typically $\mathbb{Q}$ -almost surely a constant, which is called *asymptotic growth rate* of E against $\mathbb{Q}$.

- Positive asymptotic growth rate $\not\Rightarrow$ consistency.
- An e-process may be consistent against $\mathbb{Q}$ without having positive asymptotic growth rate under $\mathbb{Q}$.

Asymptotic Growth Rate vs. Consistency

Under the null $\mathbb{P} \in \mathcal{P}$:

- For any e-process (E_t) , $\mathbb{E}^{\mathbb{P}}[E_t] \leq 1$ for all t (optional stopping)
- By Jensen and supermartingale convergence:

$$\mathbb{E}^{\mathbb{P}}[\log E_t] \leq 0 \quad \Rightarrow \quad \text{asymptotic growth rate under } \mathbb{P} \leq 0.$$

- $\Rightarrow$ under the null, E_t cannot grow on average.

Asymptotic Growth Rate vs. Consistency

Under an alternative Q :

- If $\liminf_{t \rightarrow \infty} \frac{1}{t} \log E_t > 0$ (a.s.), then $E_t \rightarrow \infty$ a.s. under Q .
- Thus $P_Q(\text{reject eventually}) = 1 \Rightarrow$ **consistency**.

Positive growth rate $\Rightarrow$ consistency.

But not conversely:

- Consistency only requires $E_t \rightarrow \infty$, not exponential growth.
- Examples: $E_t = \log(t+1)$ or $E_t = t^\beta$ ($\beta > 0$):

$$E_t \rightarrow \infty \quad \text{but} \quad \frac{1}{t} \log E_t \rightarrow 0.$$

- Hence consistency $\nRightarrow$ positive growth rate.

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Likelihood-ratio e-processes

Simple null $\mathbb{P}$ vs simple alternative $\mathbb{Q}$

Define $M_0 = 1$ and for $t \geq 1$,

$$M_t = \prod_{i=1}^t \frac{d\mathbb{Q}}{d\mathbb{P}}(X_i).$$

Then M is both an e-process and a test martingale adapted to the filtration generated by the iid data $\{X_1, X_2, \dots\}$.

Log-optimality of LR

Theorem (The likelihood ratio process is log-optimal)

Suppose that for testing $\mathbb{P}$ versus $\mathbb{Q}$, the LR process M exists for any n . Let $M' = (M'_t)$ be any e-process for $\mathbb{P}$. For any $\mathbb{F}$ -stopping time τ ,

$$\mathbb{E}^{\mathbb{Q}}[\log M_{\tau}] \geq \mathbb{E}^{\mathbb{Q}}[\log M'_{\tau}].$$

Interpretation.

- Among all e-processes valid under $\mathbb{P}$, the LR process M is *log-optimal* under $\mathbb{Q}$.
- It collects evidence against $\mathbb{P}$ (in log-scale) as efficiently as possible, even with optional stopping.
- Proof strategy: Reduce a stopping-time problem to a fixed σ -algebra problem by proving $\mathbb{Q} \ll \mathbb{P}$ on $\mathcal{F}_{\tau}$, then apply the fixed-time “LR maximizes expected log-evidence” result to M_{τ} .

Asymptotic log-optimality

Proposition (Asymptotic growth)

Assume (X_t) i.i.d. under $\mathbb{Q}$. Let $M_t^* = \prod_{i=1}^t \frac{d\mathbb{Q}}{d\mathbb{P}}(X_i)$ and $D := \text{KL}(\mathbb{Q} \parallel \mathbb{P})$.

- **Law of large numbers:**

$$\frac{1}{t} \log M_t^* \xrightarrow[t \rightarrow \infty]{a.s. \ \mathbb{Q}} \mathbb{E}^{\mathbb{Q}} \left[\log \frac{d\mathbb{Q}}{d\mathbb{P}}(X) \right] = D > 0.$$

- **Asymptotic optimality:** For any e-process M for $\mathbb{P}$,

$$\limsup_{t \rightarrow \infty} \frac{1}{t} \log M_t \leq D \quad \mathbb{Q}\text{-a.s.}$$

- M^* is consistent with asymptotic growth rate equal to D .
- Any other e-process have a slower asymptotic growth rate.

Composite alternatives $\mathcal{Q}$

Definition (Asymptotically log-optimal against $\mathcal{Q}$)

An e-process M is asymptotically log-optimal for $\mathbb{P}$ against $\mathcal{Q}$ if for every $\mathbb{Q} \in \mathcal{Q}$, we have

$$\liminf_{t \rightarrow \infty} \frac{1}{t} \left(\log M_t - \log M_t^{\mathbb{Q}} \right) \geq 0 \quad \text{in } L^1 - \text{convergence under } \mathbb{Q},$$

where $M^{\mathbb{Q}}$ is the likelihood ratio process of $\mathbb{Q}$ to $\mathbb{P}$.

Interpretation:

- M achieves at least the same **long-run** log-growth rate as the ideal LR process.
- Allows comparison when $\mathbb{Q}$ is unknown — M may depend on data (e.g. plug-in, mixture).

Asymptotic Log-Optimality against composite $\mathcal{Q}$:

Set up: Parametric families of alternatives.

- Null model: $\mathbb{P}_{\theta_0}$; alternative model: $\mathcal{Q}_{\theta}$, $\theta \in \Theta_1$ which does not include Θ_0 .
- True data-generating law Q_{θ^*} with parameter $\theta^* \in \Theta_1$.
- Assume i.i.d. observations $(X_t)_{t \geq 1}$ and filtration $\mathcal{F}_t = \sigma(X_1, \dots, X_t)$.
- Define the likelihood-ratio e-process as

$$M_t := \prod_{i=1}^t \frac{d\mathcal{Q}_{\theta_i}(X_i)}{d\mathbb{P}_{\theta_0}(X_i)}, \quad M_0 = 1,$$

where θ_i is measurable to $\sigma(X_j : j \in [i - 1])$ for each $i \in \mathbb{N}$.

Asymptotic Log-Optimality of Plug-in e-processes

Proposition (Asymptotic Log-Optimality of Plug-in e-processes)

For testing $\mathbb{P}$ against $\mathcal{Q} = \{\mathbb{Q}_\theta : \theta \in \Theta_1\}$, suppose that for each x , the log likelihood $\theta \mapsto \log \ell(x; \theta)$ is concave, and it has a gradient denoted by $\eta(x; \theta)$. If each $\theta \in \Theta_1$, $\mathbb{E}^{\mathbb{Q}_\theta}[\|\theta_i - \theta\|^2] \rightarrow 0$ as $i \rightarrow \infty$ and

$$\sup_{i \in \mathbb{N}} \mathbb{E}^{\mathbb{Q}_\theta}[\|\eta(X_i; \theta_i)\|^2] < \infty,$$

then (M_t) is asymptotically log-optimal.

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Testing as betting

- View a test (super)martingale/e-process as the *wealth* of a bettor wagering against $\mathcal{P}$.
- At each time t , choose a nonnegative *betting score* S_t whose conditional mean under $\mathbb{P} \in \mathcal{P}$ is ≤ 1 .
- Wealth/evidence multiplies: $W_t = \prod_{i=1}^t S_i$, with $W_0 = 1$.
- Testing rule: $\varphi_t := \mathbf{1}\{W_t \geq 1/\alpha\}$ (anytime).

Simple null

1. Initialize wealth $W_0 = 1$.
2. For $t = 1, 2, \dots$:
 - 2.1 Declare a bet $S_t : \mathcal{X} \rightarrow [0, \infty)$ with $\mathbb{E}^{\mathbb{P}}[S_t(X) \mid \mathcal{F}_{t-1}] \leq 1$ (for the fixed $\mathbb{P}$).
 - 2.2 Observe X_t .
 - 2.3 Update wealth: $W_t = W_{t-1} S_t(X_t)$.

Result: (W_t) is an e-process/test supermartingale for $\mathbb{P}$.

Composite null

1. For each $\mathbb{P} \in \mathcal{P}$, set $W_0^{\mathbb{P}} = 1$.
2. For $t = 1, 2, \dots$:
 - 2.1 For each $\mathbb{P} \in \mathcal{P}$, pick $S_t^{\mathbb{P}}$ with $\mathbb{E}^{\mathbb{P}}[S_t^{\mathbb{P}}(X) \mid \mathcal{F}_{t-1}] \leq 1$.
 - 2.2 Observe X_t .
 - 2.3 Update: $W_t^{\mathbb{P}} = W_{t-1}^{\mathbb{P}} S_t^{\mathbb{P}}(X_t)$ for all $\mathbb{P}$.
 - 2.4 Overall wealth: $W_t := \inf_{\mathbb{P} \in \mathcal{P}} W_t^{\mathbb{P}}$.

Result: (W_t) is an e-process for $\mathcal{P}$.

Connection to e-processes and tests

- Betting wealth $W_t \equiv$ e-process for $\mathcal{P}$.
- Thresholding wealth at $1/\alpha$ yields a valid sequential test.
- For composite $\mathcal{P}$, taking $W_t = \inf_{\mathbb{P} \in \mathcal{P}} W_t^{\mathbb{P}}$ maintains the e-process property.

Interpretation: “Evidence = wealth”: larger W_t indicates stronger evidence against the null.

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